

Math100C - WK7

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1.

Take field extensions $\mathbb{F}_t/\mathbb{F}_q$ of finite fields. These fields must have the same character, namely of some prime p . That is, one really has field extensions

$$\mathbb{F}_t/\mathbb{F}_q/\mathbb{F}_p$$

It was shown in week 6 that the primitive element theorem holds for finite extensions of $\mathbb{F}_p$, so there is some $\alpha \in \mathbb{F}_t$ such that $\mathbb{F}_t = \mathbb{F}_p(\alpha)$. Then

$$\mathbb{F}_t = \mathbb{F}_q(\alpha);$$

By definition, $\mathbb{F}_q(\alpha) \subset \mathbb{F}_t$. To get containment in the other direction, $\mathbb{F}_q(\alpha)$ contains $\mathbb{F}_p$ and α , i.e., $\mathbb{F}_t = \mathbb{F}_p(\alpha) \subset \mathbb{F}_q(\alpha)$.

2.

The backward direction follows because the Frobenius map is an automorphism of F :

$$c(u+v)^{p^n} = cu^{p^n} + cv^{p^n},$$

hence if each term of f has degree $c_i x^{p^{n_i}}$, then $u \mapsto f(u)$ is a homomorphism on F .

3.

Recall the fundamental theorem for finitely generated modules over a PID R : if M is such a module, there exist a number r, m and nonunit elements $a_1 \mid a_2 \mid \dots \mid a_m$ such that

$$M \cong R^{\oplus r} \oplus R/\langle a_1 \rangle \oplus R/\langle a_2 \rangle \oplus \dots \oplus R/\langle a_m \rangle$$

By definition, finitely generated free modules over R are those of the form $R^{\oplus r}$. Suppose M is finitely generated. If M is free, then because R is a PID and thus particularly an integral domain, $M = R^{\oplus r}$ has no torsion elements.

Vice versa, if M is torsion-free, then all the $R/\langle a_i \rangle$ factors need to be the zero module, for $R/\langle a_i \rangle$ as a R -module has torsion elements: namely, for any $b + \langle a_i \rangle \in R/\langle a_i \rangle$, $a_i \cdot (b + \langle a_i \rangle) = 0$ in $R/\langle a_i \rangle$. So $M \cong R^{\oplus r}$, which proves M is a free R -module.

If R is any ring, recall that considering R as a module over itself, the submodules of R are precisely the ideals in R . Let $R = \mathbb{C}[x, y]$. Consider the ideal $\langle x, y \rangle \triangleleft \mathbb{C}[x]$ as an R -module. It is clearly torsion-free because R is an integral domain. It is also intuitively clear that it is a free-module, though I can't find a proof...

4.

I saw this proof, which is also the proof intended by Kiran, while going through Dummit and Foote. Suppose $\Phi_n(x) = f(x)g(x)$ is a factorization of Φ_n , where f is irreducible over $\mathbb{Q}$. Fix a primitive n th root of unity $\zeta \in \mathbb{C}$ that is a root of f (i.e., f is the minimal polynomial for ζ over $\mathbb{Q}$; all roots of f are a primitive n th root of unity because they are, by construction, the roots of $\Phi_n(x)$), and take any prime p not dividing n . Suppose ζ^p is a root of g . Then ζ is a root of $g(x^p)$, so $g(x^p) = f(x)h(x)$ for some $h \in \mathbb{Z}[x]$ (by Gauss Lemma). Over $\mathbb{F}_p$, $\bar{g}(x^p) = \bar{g}(x)$ gives

$$\bar{g}(x) = \bar{f}(x)\bar{h}(x)$$

Hence over $\mathbb{F}_p$, $\bar{\Phi}_n(x)$ has a repeated factor; since $\Phi_n(x)$ is a factor of $x^n - 1$ in $\mathbb{Z}[x]$, it follows $x^n - 1$ has a repeated factor over $\mathbb{F}_p$. This contradicts the obvious observation that $x^n - 1$ doesn't have repeated roots over $\mathbb{F}_p$ since its derivative is nx^{n-1} , which isn't 0 since $p \nmid n$; $x^n - 1$ and its derivative have no common factor, so $x^n - 1$ can't have a repeated root. This shows that ζ^p is not a root of g ; since ζ^p is a root of $\Phi_n(x)$, it follows ζ^p is a root of f .

Let a be coprime to n . Then $a = p_1 \cdots p_k$ for primes $p_i \nmid n$. Since the argument in the paragraph above was for an arbitrary root ζ of f and any prime not dividing n , ζ^a is a root of f ; the conclusion is

$$\zeta^a \text{ is a root of } f \text{ for every } a \text{ coprime to } n$$

i.e., every primitive n th root of unity is a root of f . This forces $f = \Phi_n$; $\Phi_n(x)$ is irreducible.